vectors

A matrix with only one column is called a vector or a column vector.

Linear Combinations

Suppose we are given vectors $\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_p$ in $\mathbb{R}^n$, and scalars $c_1, ..., c_p$. The vector

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

is called a linear combination of $\mathbf{v}_1, ..., \mathbf{v}_p$ with weights $c_1, ..., c_p$.

Example. Let

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ -1 \\ 2 \\ -3 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ -1 \\ -2 \\ 1 \end{bmatrix}$$

and $c_1 = -1, c_2 = 2, c_3 = -2$. Then

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + c_3\mathbf{v}_3 = (-1)\begin{bmatrix} 3 \\ -1 \\ 2 \\ -3 \end{bmatrix} + 2\begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix} + (-2)\begin{bmatrix} 0 \\ -1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 0 \\ 5 \end{bmatrix}$$

is a linear combination of $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$.

A vector equation is an equation of the form

$$x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$$

where $x_1, \ldots, x_p$ are scalars and $\mathbf{b}, \mathbf{a}_1, \ldots, \mathbf{a}_p \in \mathbb{R}^n$.

$$x_1\begin{bmatrix}1\\-2\\1\end{bmatrix} + x_2\begin{bmatrix}-2\\1\\1\end{bmatrix} + x_3\begin{bmatrix}3\\-3\\0\end{bmatrix} = \begin{bmatrix}1\\3\\-1\end{bmatrix} \rightsquigarrow \begin{bmatrix}x_1 - 2x_2 + 3x_3\\-2x_1 + x_2 - 3x_3\\x_1 + x_2\end{bmatrix} = \begin{bmatrix}1\\3\\-1\end{bmatrix}$$

Relation between vector equations and linear systems

- The vector equation

  $$x_1\mathbf{a}_1 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$$

  has the same solution set as the linear system whose augmented matrix is

  $$\left[\begin{array}{cccc|c}\mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p & \mathbf{b}\end{array}\right].$$

Summary

Given vectors $\mathbf{b}$ and $\mathbf{a}_1, \ldots, \mathbf{a}_p$ in $\mathbb{R}^n$, the following statements are equivalent (i.e., if any of them is true, then the rest are true):

- $\mathbf{b}$ is a linear combination of $\mathbf{a_1}, ..., \mathbf{a_p}$.
- the linear system with augmented matrix

$$\left[\begin{array}{ccc|c} \mathbf{a}_1 & \cdots & \mathbf{a}_p & \mathbf{b} \end{array}\right]$$

  is consistent.

Matrix Product : Special Case

Suppose $A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$ is a $1 \times n$ matrix and $\mathbf{v} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ is an $n \times 1$ vector.Then the product $A\mathbf{v}$ is defind as follows :

$$A\mathbf{v} = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \begin{bmatrix} a_1 b_1 + a_2 b_2 + \cdots + a_n b_n \end{bmatrix}$$

The result is a 1×1 matrix.

Example.

$$\begin{bmatrix} 1 & -1 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} (1\times 2) + ((-1)\times(-3)) + (0\times 1) + (2\times 1) \end{bmatrix} = \begin{bmatrix} 7 \end{bmatrix}$$

Example. The product of a 1×5 matrix and a 3×1 vector does not make sense.

$$\begin{bmatrix} * & * & * & * & * \end{bmatrix} \begin{bmatrix} * \\ * \\ * \end{bmatrix} = \text{☹}$$

• The product $A\mathbf{v}$ makes sense only if the number of columns of A is the same as the number of rows of $\mathbf{v}$.

Matrix Product : General Case

Given an $m \times n$ matrix A and an $n \times 1$ vector $\mathbf{v}$, the product $A\mathbf{v}$ is defined as follows :

$$A\mathbf{v} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

So, the result is an $m \times 1$ vector.

product of $m \times n$ and $n \times 1 \rightsquigarrow m \times 1$

Example.

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & -1 & -2 & 1 \\ 2 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} + (-3) \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ -1 \end{bmatrix}$$

Matrix Product : General Case

Given an $m \times n$ matrix A and an $n \times 1$ vector $\mathbf{v}$, the product $A\mathbf{v}$ is defined as follows :

$$A\mathbf{v} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n$$

So, the result is an $m \times 1$ vector.

product of $m \times n$ and $n \times 1$ $\rightsquigarrow$ $m \times 1$

Example.

$$\begin{bmatrix} 1 & 2 & 3 & 1 & 0 \\ -2 & 1 & -1 & 3 & 1 \\ 2 & 7 & -1 & -1 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} = ?$$

This product does not make sense. ☹

Another way of computing the product...

A : $m \times n$ matrix.

$\mathbf{v}$: $n \times 1$ vector,

$A\mathbf{v}$: $m \times 1$ vector.

The k-th entry $A\mathbf{v}$ = product of the k-th row of A with $\mathbf{v}$.

$$\begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & -1 & -2 & 1 \\ 2 & -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1\times 2 + (-1)\times 1 + 2\times(-3) + 0\times(-1) \\ 0\times 2 + (-1)\times 1 + (-2)\times(-3) + 1\times(-1) \\ 2\times 2 + (-1)\times 1 + 1\times(-3) + 1\times(-1) \end{bmatrix}$$

$$= \begin{bmatrix} -5 \\ 4 \\ -1 \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} -3 \\ 0 \\ 1 \\ 5 \end{bmatrix} + (-3) \begin{bmatrix} 4 \\ 1 \\ -2 \\ -1 \end{bmatrix} = \begin{bmatrix} -13 \\ 1 \\ 7 \\ 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 4 \\ 2 & 0 & 1 \\ 0 & 1 & -2 \\ 0 & 5 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \times 2 + (-3) \times 1 + 4 \times (-3) \\ 2 \times 2 + 0 \times 1 + 1 \times (-3) \\ 0 \times 2 + 1 \times 1 + (-2) \times (-3) \\ 0 \times 2 + 5 \times 1 + (-1) \times (-3) \end{bmatrix} = \begin{bmatrix} -13 \\ 1 \\ 7 \\ 8 \end{bmatrix}$$

Matrix product and linear combination

Suppose $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ are vectors in $\mathbb{R}^3$.

We can write the expression $2\mathbf{a} - 3\mathbf{b} + \mathbf{c} - \mathbf{d}$ using matrix products as follows :

$$2\mathbf{a} - 3\mathbf{b} + \mathbf{c} - \mathbf{d} = \begin{bmatrix} \mathbf{a} & \mathbf{b} & \mathbf{c} & \mathbf{d} \end{bmatrix} \begin{bmatrix} 2 \\ -3 \\ 1 \\ -1 \end{bmatrix}$$

The above expression is a product of a 3×4 matrix with a 4×1 column vector.

Matrix Equation Form of a Linear System

Suppose we are given a linear system : $\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$

1. We can write it in matrix form as follows :

$$\begin{bmatrix} -x_1 & +2x_2 & +x_3 \\ 2x_1 & -4x_2 & \\ -x_1 & +2x_2 & +3x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$$

2. Then we "separate" the variables :

$$\begin{bmatrix} -x_1 \\ 2x_1 \\ -x_1 \end{bmatrix} + \begin{bmatrix} 2x_2 \\ -4x_2 \\ 2x_2 \end{bmatrix} + \begin{bmatrix} x_3 \\ 0 \\ 3x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$$

3. We can now rewrite it as:

$$x_1 \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ -4 \\ 2 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix} \rightsquigarrow \begin{bmatrix} -1 & 2 & 1 \\ 2 & -4 & 0 \\ -1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}$$

Matrix equation form of the linear system.

Therefore for the linear system

$$\begin{cases} -x_1 & +2x_2 & +x_3 & = 1 \\ 2x_1 & -4x_2 & & = -4 \\ -x_1 & +2x_2 & +3x_3 & = -1 \end{cases}$$

we have obtained the following matrix equation :

$$\underbrace{\begin{bmatrix} -1 & 2 & 1 \\ 2 & -4 & 0 \\ -1 & 2 & 3 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 1 \\ -4 \\ -1 \end{bmatrix}}_{\mathbf{b}}$$

We obtain the equation $A\mathbf{x} = \mathbf{b}$ where :

- A is the coefficient matrix.
- $\mathbf{x}$ is the column of variables.
- $\mathbf{b}$ is the rightmost column of the augmented matrix.

The solution set of the linear system is the same as the solution set of the matrix equation.

Example

$$\begin{cases} y_1 & -2y_2 & & +y_4 = -2 \\ & -3y_2 & +5y_3 & -3y_4 = -6 \\ -y_1 & -7y_2 & & -y_4 = -3 \end{cases} \rightsquigarrow \begin{bmatrix} 1 & -2 & 0 & 1 \\ 0 & -3 & 5 & -3 \\ -1 & -7 & 0 & -1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} -2 \\ -6 \\ -3 \end{bmatrix}$$

Consequently...

If $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_p, \mathbf{b}$ are vectors in $\mathbb{R}^n$, then the following equations have the same solution set :

1. The linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p & \mathbf{b} \end{array}\right],$$

2. The vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$,

3. The equation $A\mathbf{x} = \mathbf{b}$ where $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$.

If $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_p, \mathbf{b}$ are vectors in $\mathbb{R}^n$, then the following statements are logically equivalent (i.e., they are either all true or all false):

1. The linear system whose augmented matrix is

$$\left[\begin{array}{cccc|c} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p & \mathbf{b} \end{array}\right]$$

   is consistent.

2. The vector equation $x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_p\mathbf{a}_p = \mathbf{b}$ has a solution.

3. $\mathbf{b}$ is a linear combination of $\mathbf{a}_1, \mathbf{a}_2, ..., \mathbf{a}_p$.

4. $\mathbf{b}$ is in $\text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_p\}$.

5. The equation $A\mathbf{x} = \mathbf{b}$ where $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_p \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix}$

   has a solution.

Example

Suppose that:

- A is an $m \times n$ matrix,
- $\mathbf{x}$ is an $n \times 1$ vector with variable entries (e.g, $x_1, ..., x_n$).

Question : When does $A\mathbf{x} = \mathbf{b}$ have a solution for every $\mathbf{b}$ in $\mathbb{R}^m$?

Answer : When every row of A contains a pivot position.

(Because then in the R.E.F. of $\left[\, A \mid \mathbf{b} \,\right]$ the rightmost column can never be a pivot column.)

Guaranteed Existence of Solution

Suppose $A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$ is an $m \times n$ matrix. Then the following statements are logically equivalent (i.e., they are either all true or all false) :

1. For every vector $\mathbf{b}$ in $\mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
2. Every vector $\mathbf{b}$ in $\mathbb{R}^m$ is a linear combination of $\mathbf{a}_1, ..., \mathbf{a}_n$.
3. The columns of A span $\mathbb{R}^m$, i.e., $\text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\} = \mathbb{R}^m$.
4. A has a pivot position in every row.

Example

Question. Suppose $A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix}$. Does the equation $A\mathbf{x} = \mathbf{b}$ have a solution for every $\mathbf{b}$ in $\mathbb{R}^3$?

Answer. No, because the R.E.F. of A is:

$$\begin{bmatrix} 1 & 0 & -\frac{4}{3} \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and the last row does not contain a pivot position.

Application : Network Flow

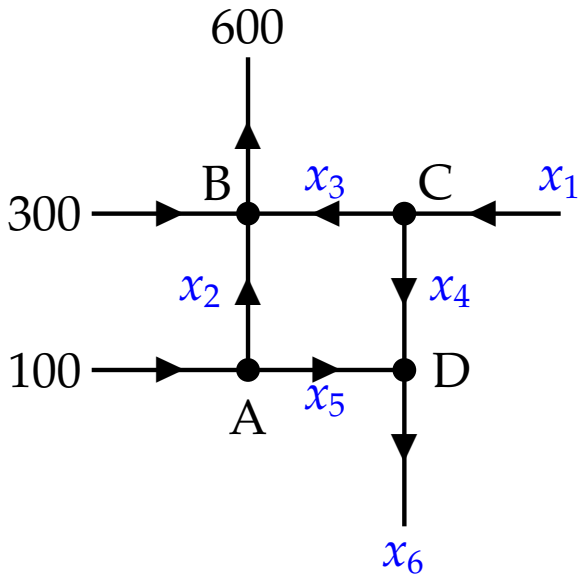


Applications:

1. Traffic modelling,
2. Electrical circuits,
3. Input/output analysis of economical models

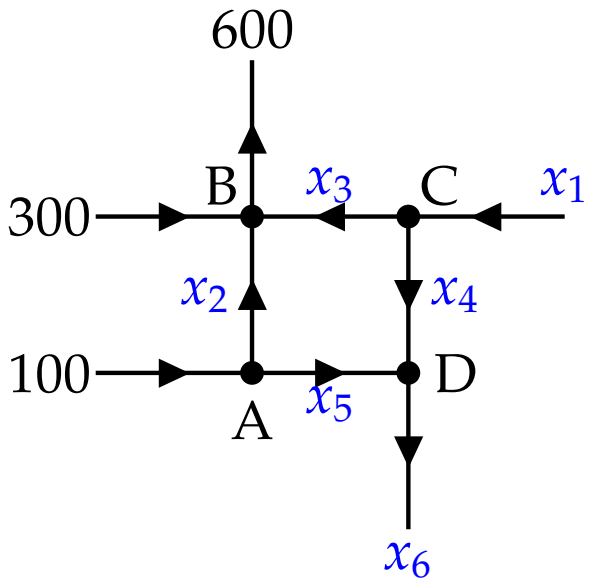
Application : Network Flow (cont.)

Suppose that the traffic flow of a city is depicted by the following diagram :



- The numbers and variables represent the number of cars per hour. Therefore $x_1, ..., x_5$ should be non-negative.
- The arrows indicate that the streets are one way.

Problem. Determine the general flow pattern.



Solution. We write equations according to two rules:

1. We should have total flow in = total flow out.

$$300 + x_1 + 100 = 600 + x_6$$

2. At each node we should have flow in = flow out.

(A)	100	$=$	$x_2 + x_5$
(B)	$300 + x_2 + x_3$	$=$	600
(C)	x_1	$=$	$x_3 + x_4$
(D)	$x_4 + x_5$	$=$	x_6

Therefore we obtain a linear system :

$$\begin{cases} x_1 & & & & & -x_6 & = 200 \\ x_1 & & -x_3 & -x_4 & & & = 0 \\ & x_2 & +x_3 & & & & = 300 \\ & x_2 & & & +x_5 & & = 100 \\ & & & x_4 & +x_5 & -x_6 & = 0 \end{cases}$$

Augmented matrix of the system is :

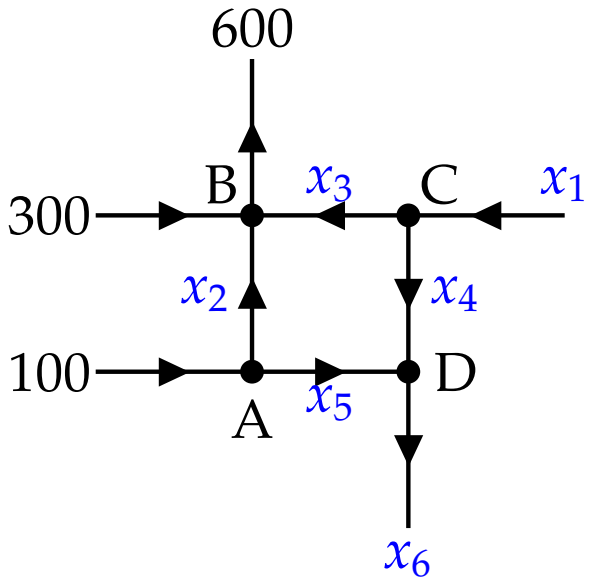
$$\left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 0 & -1 & 200 \\ 1 & 0 & -1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 300 \\ 0 & 1 & 0 & 0 & 1 & 0 & 100 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \end{array}\right] \xrightarrow{\text{Row Reduction}} \left[\begin{array}{cccccc|c} 1 & 0 & 0 & 0 & 0 & -1 & 200 \\ 0 & 1 & 0 & 0 & 1 & 0 & 100 \\ 0 & 0 & 1 & 0 & -1 & 0 & 200 \\ 0 & 0 & 0 & 1 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

and after calculating the R.E.F. we realize that :

- x_1, x_2, x_3, x_4 are basic.
- x_5, x_6 are free.

And the parametric solution is : $\begin{cases} x_1 = x_6 + 200 \\ x_2 = 100 - x_5 \\ x_3 = x_5 + 200 \\ x_4 = -x_5 + x_6 \\ x_5 : \text{free} \\ x_6 : \text{free} \end{cases}$

This solution is called the general flow pattern



General flow pattern :
$$\begin{cases} x_1 = x_6 + 200 \\ x_2 = 100 - x_5 \\ x_3 = x_5 + 200 \\ x_4 = -x_5 + x_6 \\ x_5 : \text{free} \\ x_6 : \text{free} \end{cases}$$

Question. What is the maximum possible amount of flow from node A to node B?

Solution. Since $x_2 = 100 - x_5$ and $x_5 \geq 0$, it follows that $x_2 \leq 100$.